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## 12.8. POLYMERS VERSUS FREQUENCY

**Table 12.8** shows the parametric fits to permittivity for various polymers. The functional forms are shown at the beginning of the table. Functions are those causal functions that have been derived in Chaps. 2 and 3 for dielectric and magnetic media. Samples were measured in waveguide, focused beam, and Fabry–Perot systems and data were integrated for processing. Other data are shown when available. These include density, measured frequency span serving as basis for fits, tabulated information, and chemical and/or physical composition. In the tables exponentials are often shown as  $bE$ —which means  $b \cdot 10^{-a}$ . Thus,  $-2E-004-7.6E-003j$  is equal to  $-0.0002-j0.0076$ . **Figure 12.11** indicates typical fits to the measurement: the graph at the top shows a very low loss polymer, Rexolite. Measured  $\epsilon_{imag}$  shows unphysical rapid ripples in the imaginary permittivity. These are smoothed by a fit to the causal function. The lower graph of the figure is for a higher loss epoxy. In that case the need for smoothing is not strong.

**Table 12.8** Polymers versus Frequency

| Sample ID                                                                                                        |            |            |            |            |             |            |
|------------------------------------------------------------------------------------------------------------------|------------|------------|------------|------------|-------------|------------|
| $\epsilon(f) = B + 2Cf^D + G\{1 - J*(f - H)^2 - j2If\}^{-1}$                                                     |            |            |            |            |             |            |
| B                                                                                                                | C          | D          | G          | H          | I           | J          |
| Bismaleimide BMI F650, 1.27 g/cc (18–60 GHz)                                                                     |            |            |            |            |             |            |
| 0.5471+                                                                                                          | 0.6184+    | 0.1048–    | 1.7022–    | 1.53E-002+ | –1.5E-003   | 2E-004     |
| 3.87E-002j                                                                                                       | 3.24E-002j | 7.1E-003j  | 4.33E-002j | 2.2E-003j  | +1.62E-002j |            |
| Bismaleimide BMI F650, 1.27 g/cc (18–100 GHz)                                                                    |            |            |            |            |             |            |
| 0.7982+                                                                                                          | 0.7982+    | 5.4E-003+  | 0.7987+    | 5E-004+    | –1.9E-003+  | 5.91E-005+ |
| 2.17E-002j                                                                                                       | 2.17E-002j | 3.8E-003j  | 2.16E-002j | 5E-004j    | 2.6E-003j   | 2.91E-005j |
| Celazole polybenzamidazole PBI (75–100 GHz), 1.3 g/cc (nominal $\epsilon_r = 4.2$ at 1 MHz, Boedeker data sheet) |            |            |            |            |             |            |
| 0.6934+                                                                                                          | 0.6812+    | –0.1589+   | 0.6014–    | 4.6E-003+  | –1E-003–    | –1E-004    |
| 0.1502j                                                                                                          | 0.3522j    | 1.96E-002j | 4.5E-003j  | 6.7E-003j  | 7.8E-003j   |            |
| Cyanate ester 561, (20–60 GHz), 1.22 g/cc                                                                        |            |            |            |            |             |            |
| 0.707+                                                                                                           | 0.707+     | 4E-003–    | 0.7075+    | 7E-004–    | 9E-004+     | 1E-004     |
| 1.3E-003j                                                                                                        | 1.3E-003j  | 9E-004j    | 1.1E-003j  | 2E-004j    | 3.8E-003j   |            |
| Cyanate ester-2 (30–100 GHz), 1.22 g/cc                                                                          |            |            |            |            |             |            |

Sample ID

$$\epsilon(f) = B + 2Cf^D + G\{1 - J*(f - H)^2 - j2If\}^{-1}$$

|                                                                                                                                                |            |             |            |            |            |             |
|------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|------------|------------|------------|-------------|
| 0.6469+                                                                                                                                        | 0.6796+    | 8.81E-002-  | 0.4457+    | 1.06E-002- | -3.4E-003- | -6E-004     |
| 5.46E-002j                                                                                                                                     | 4.16E-002j | 1.67E-002j  | 8.11E-002j | 5.5E-003j  | 1.22E-002j |             |
| Epoxy, EAB (2–100 GHz), 1.24 g/cc                                                                                                              |            |             |            |            |            |             |
| 1.0702+                                                                                                                                        | 0.8609-    | -0.5203+    | 0.8918-    | -2E-003+   | 1E-004-    | -4.85E-005+ |
| 5.32E-002j                                                                                                                                     | 2.38E-002j | 0.1529j     | 9.54E-002j | 1.7E-003j  | 4.8E-003j  | 5.91E-008j  |
| Polyester F141 (18–60 GHz), 1.38 g/cc                                                                                                          |            |             |            |            |            |             |
| 0.7543+                                                                                                                                        | 0.7538+    | 1.1E-003+   | 0.7653+    | 7.1E-003+  | -6.3E-003+ | 2E-004+     |
| 3.25E-002j                                                                                                                                     | 3.23E-002j | 4.7E-003j   | 3.51E-002j | 3.5E-003j  | 6.6E-003j  | 2E-004j     |
| Epoxy F161, 1.243 g/cc (18–100 GHz)                                                                                                            |            |             |            |            |            |             |
| 0.8804+                                                                                                                                        | 0.9002+    | -6.88E-002- | 0.8758+    | -6E-004+   | -1.5E-003j | -1.12E-005+ |
| 2.21E-002j                                                                                                                                     | 2.34E-002j | 1.24E-002j  | 2.13E-002j | 7E-004j    |            | 1.61E-007j  |
| 2555 Meltbond adhesive (20– 60 GHz)                                                                                                            |            |             |            |            |            |             |
| 0.7346+                                                                                                                                        | 0.7345+    | 1.2E-003+   | 0.7355+    | 4E-004     | -2E-004+   | 9.70E-005+  |
| 4.1E-003j                                                                                                                                      | 4.1E-003j  | 1E-004j     | 4.1E-003j  |            | 3.5E-003j  | 5.53E-006j  |
| Plexiglass acrylic, poly(methyl methacrylate) (PMMA), 1.2 g/cc (26–100 GHz) (nominal $\epsilon_r \sim 2.5$ )                                   |            |             |            |            |            |             |
| 0.6719+                                                                                                                                        | 0.6724+    | 1.85E-002+  | 0.6692+    | 5.2E-003+  | -1E-004+   | 8.54E-005+  |
| 2E-003j                                                                                                                                        | 2E-003j    | 6E-004j     | 2E-003j    | 1E-004j    | 6E-003j    | 2.25E-006j  |
| Polyester, F148 (18–100 GHz), about 1.4g/cc                                                                                                    |            |             |            |            |            |             |
| 0.9813+                                                                                                                                        | 0.931-     | 6.8E-003+   | 7E-003+    | 3.56E-002- | -2.4E-003- | -9E-004+    |
| 8.28E-002j                                                                                                                                     | 5.15E-002j | 2.73E-002j  | 1.75E-002j | 7E-003j    | 2.89E-002j | 1E-004j     |
| 1422, Cross-linked polystyrene 1.05 g/cc (4–100 GHz) (nominal $\epsilon_r = 2.53$ , 1–500 GHz, Rexolite Data Specs. complast.com)              |            |             |            |            |            |             |
| 0.5914+                                                                                                                                        | 0.586+     | 2.24E-002+  | 0.7488-    | -2E-004+   | -1.4E-003+ | 2.74E-005+  |
| 5.09E-002j                                                                                                                                     | 4.83E-002j | 6.8E-003j   | 0.152j     | 1.3E-003j  | 2.1E-003j  | 1.78E-005j  |
| 1000 Polyetherimide (2–60 GHz), 1.28 g/cc (nominal $\epsilon_r = 3.15$ , $\tan \delta = .0013$ , Boedeker Ultem Polyetherimide Specifications) |            |             |            |            |            |             |
| 0.7855+                                                                                                                                        | 0.7872+    | -2.93E-002- | 0.7851+    | 8E-004+    | 1E-004-    | -4.51E-005- |
| 9E-004j                                                                                                                                        | 8E-004j    | 7E-004j     | 1E-003j    | 1E-004j    | 2.7E-003j  | 2.27E-006j  |

Sample ID

$$\epsilon(f) = B + 2Cf^D + G\{1 - J*(f - H)^2 - j2If\}^{-1}$$

Polyimide bismaleimide F178 (18–100 GHz), 1.297 g/cc

|                      |                       |                      |                    |                        |                        |                          |
|----------------------|-----------------------|----------------------|--------------------|------------------------|------------------------|--------------------------|
| 0.504–<br>4.24E-002j | 0.5854–<br>4.11E-002j | 0.1004+<br>5.5E-003j | 1.6253+<br>0.1025j | 8.6E-003–<br>1.7E-003j | 2.1E-003+<br>9.6E-003j | 9.82E-005–<br>1.99E-005j |
|----------------------|-----------------------|----------------------|--------------------|------------------------|------------------------|--------------------------|

Polyethelene (8–40 GHz), 0.93–0.94 g/cc (nominal  $\epsilon_r = 2.25$ –2.35)

|                    |                      |                     |                    |                       |                        |                           |
|--------------------|----------------------|---------------------|--------------------|-----------------------|------------------------|---------------------------|
| 0.6329+<br>7E-004j | 1.0517–<br>2.1E-003j | –0.2925–<br>2E-004j | 0.5377+<br>3E-003j | –2.2E-003+<br>7E-004j | –1E-004–<br>1.11E-002j | –2.67E-004+<br>2.19E-006j |
|--------------------|----------------------|---------------------|--------------------|-----------------------|------------------------|---------------------------|

Polytetrafluoroethelene (10–100 GHz), 2.13–2.19 g/cc (nominal  $\epsilon_r = 2.01$ )

|                    |                    |                     |                    |                    |            |                           |
|--------------------|--------------------|---------------------|--------------------|--------------------|------------|---------------------------|
| 0.5635+<br>2E-004j | 0.6481+<br>6E-004j | –0.1309+<br>1E-004j | 0.5095–<br>7E-004j | 9E-004+<br>2E-004j | –3.4E-003j | –3.40E-005+<br>1.74E-007j |
|--------------------|--------------------|---------------------|--------------------|--------------------|------------|---------------------------|

Polyamide 7 (frequency averaged value between 10 and 16 GHz), 1.15–1.25 g/cc

3.1+3E-003j

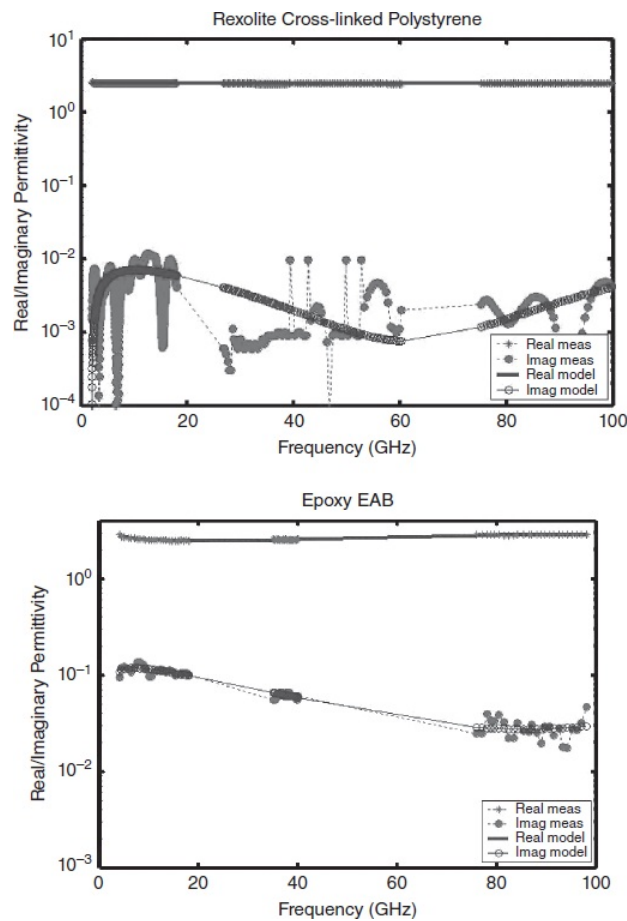
$$\epsilon(f) = B + 2Cf^D + Ef^J + G\{1 - J*(f - H)^2 - j2If\}^{-1}$$

B C D E F G H I J

7058 Epoxy with about 40%/volume 52- $\mu$ m silica hollow spheres, 3–5 $\mu$ m walls (2–18 GHz)

|                       |                    |                     |                    |                     |                       |                        |                               |                               |
|-----------------------|--------------------|---------------------|--------------------|---------------------|-----------------------|------------------------|-------------------------------|-------------------------------|
| 0.7275+<br>7.58E-002j | 2.4321+<br>0.1323j | –3.8007–<br>0.9919j | 1.6261+<br>0.2312j | –3.3603+<br>1.9301j | 0.9885–<br>5.86E-002j | –9E-003<br>+1.07E-002j | –2E-<br>004–<br>2.3E-<br>003j | –1.62E-<br>004+<br>1.21E-005j |
|-----------------------|--------------------|---------------------|--------------------|---------------------|-----------------------|------------------------|-------------------------------|-------------------------------|

**Figure 12.11** The upper figure illustrates how functional fits can minimize unphysical data characteristics. This often occurs for low loss materials. The lower figure illustrates a higher loss measurement and functional fit.



**Disclaimer:** Data are results obtained by the author and colleagues on a small number of samples purchased or obtained as free samples from suppliers. Data are not "guaranteed" by the manufacturer or author. Variations in properties may be observed and users of the products should verify properties. All material properties are guides for engineering design. Materials prepared in the laboratory should be considered as experimental and also guides for engineering design.